

VIBHA BHAVIKATTI

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EDUCATION

San Jose State University, San Jose, California
Master of Science in Computer Science

August 2024 - June 2026 (Spring 2026)

American University of Sharjah, Sharjah, UAE
Bachelor of Science in Computer Science, Minor in Data Science

September 2020 - June 2024

Dean's List Honoree: Fall 2020, Spring 2021, Fall 2021

Chancellor's List Honoree: Fall 2021

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Software Engineering, Distributed Systems, Machine Learning, Database Systems, Artificial Intelligence, Deep Learning, NLP, Cloud Computing, LLM, Linear Algebra, Cybersecurity, Computer Networks

SKILLS

- **Programming:** Python, Java, C++, SQL, TypeScript, JavaScript
- **Machine Learning & AI:** PyTorch, NLP, LangGraph, RAG, Agentic Systems, LLM Evaluation, Adversarial Testing, Prompt Engineering
- **Backend & Systems:** REST APIs, FastAPI, Flask, Distributed Systems, ETL Pipelines
- **Cloud & DevOps:** AWS, Docker, CI/CD, Linux
- **Databases:** PostgreSQL, MongoDB, Neo4j, Oracle
- **Tools:** Git, Testing (Cypress, Vitest), Selenium, Locust, Cursor, Claude Code, Ollama

EXPERIENCE

Scale AI

Aug 2025 - Present

Technical Advisor Intern - GenAI

- Built **evaluation pipelines** to analyze LLM outputs, identifying failure patterns, safety risks, and prompt injection vulnerabilities, and performed root-cause analysis to improve reliability and consistency across edge cases.
- Delivered **10+ high-priority evaluation tasks** under tight timelines, consistently completing rapid-turnaround projects to support model validation and deployment readiness.
- **Led technical reviews** across a **team of 4 interns**, improving evaluation quality, and test coverage.
- Developed automated workflows using **Selenium** and **Locust** to simulate real-world usage and adversarial inputs, enabling large-scale evaluation of model behavior.

Google Developer Groups (GDG) – SJSU

Technical Lead

August 2025 – May 2026

- Led AI and cloud engineering workshops for **30+ student members**, mentoring developers on scalable systems and **Google Cloud** technologies.

American University of Sharjah

Teaching Assistant

Sept 2022 – May 2024

- Mentored **100+ students** in Programming I, Programming Languages and Digital Systems across 4 semesters; provided hands-on support in lab sessions and code reviews.

PROJECTS

Prior Authorization Review Agent | Python, LangGraph, RAG, FastAPI, MCP, Claude Code

June 2026 - Present

- Building an agentic prior authorization review system using **LangGraph** with a proposer/critic/judge pipeline to evaluate clinical requests against **CMS policy documents** via RAG.
- Implemented document ingestion and retrieval pipeline over CMS policy PDFs using vector embeddings for grounded decision outputs.
- Integrated **MCP tools** for external context retrieval and designed a modular FastAPI backend with structured audit traces for each authorization decision.

AutoRedTeam: LLM Red Teaming System | Python, NVIDIA Nemotron-253B, Ollama, LLM-as-Judge

Apr 2026 - Present

- Built an autonomous red teaming system using **Nemotron-Ultra-253B** (via NVIDIA NIM) to generate adversarial prompts across jailbreak, prompt injection, data extraction, and bias attacks on a local **Llama 3.2** model.
- Implemented an **LLM-as-judge** pipeline with a rule-based safety net to prevent judge over-trust, achieving a **63.6% overall attack success rate** across 33 trials and identifying prompt injection as the most defended attack type (0% success).
- Designed a modular, config-driven system supporting model swapping and OpenAI-compatible APIs for testing different LLM endpoints.

Malware Classification & Explainability Analysis (Master's Project) | Python, PyTorch, Grad-CAM, XGBoost

Aug 2025 - May 2026

- Built an image-based malware classification pipeline across 17 families using MobileNetV2 CNNs on 8 transformations of binaries.
- Applied **explainable AI** methods such as **Grad-CAM** and quantitative metrics (Deletion/Insertion AUC, Spearman correlation) to evaluate model interpretability and stability.
- Designed multi-representation feature fusion (4096-D), achieving **0.777 accuracy** and surpassing the prior benchmark (0.75).

Neurosymbolic Compliance Engine (NEC) - Stanford TreeHacks | Python, FastAPI, Pydantic, TypeScript

February 2026

- Built a backend system combining rule-based logic and LLM reasoning for structured multi-step decision workflows.
- Used **Pydantic** models to enforce typed schemas for inputs/outputs and validate structured LLM responses; deployed a **FastAPI backend** returning decisions with full audit traces.
- Modeled regulations as a **knowledge graph** and implemented rule parsing with dependency resolution.

COMPETITIONS

KIBO Robot Programming Competition - JAXA and NASA

April 2023 – October 2023

- Simulated and developed Astrobée's laser-based ammonia leak detection and autonomous QR code scanning; final solution was executed aboard the **International Space Station (ISS)**, achieving **1st** place in the UAE and **6th** globally.